UNIT 7 - COMPLETE NOTES

Linear Equations and Inequalities

Solving linear equations in one variable, equations with radicals and absolute value, and linear inequalities shown on the number line.

What this unit covers

- **7.1** Linear equations in one variable with rational coefficients
- **7.1** Word problems - ages and numbers - turned into equations
- **7.1** Equations involving radicals, and extraneous roots
- **7.2** Absolute value and equations involving absolute value
- **7.3** Linear inequalities, the number line, and the properties of inequalities
- **7.4** Solving linear inequalities with rational coefficients

How to use these notes

Read one section at a time and work through every worked example with pen and paper before looking at the solution. Yellow boxes hold the rules most often tested in board papers. The last two pages give solved examples, board-style multiple-choice questions and short questions for self-testing.

Exam focus: Board papers regularly ask to (1) solve a linear equation with fractions, (2) solve a radical equation and reject the extraneous root, (3) solve an absolute-value equation, (4) name the property of inequality used in a step, (5) solve an inequality with rational coefficients and show it on a number line, and (6) solve a double inequality.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 7, starting at book page 173. Notes prepared by Shah Fahad.

1. Linear equations in one variable

> **Definition - Linear equation:** An equation of the form **ax + b = 0**, where a is not zero and a is a rational number, is a linear equation in one variable with rational coefficient. The highest power of the variable is 1 - there is no x^2, no square root of x, and no variable in a denominator.

The book's examples (book section 7.1.1): 2x + 3 = 0, (5/2)y - 4 = 0, 5x - 15 = 2x + 3 and (1/3)x - 2/3 = 4 are all linear equations in one variable. A value of the variable that makes the left-hand side (LHS) equal to the right-hand side (RHS) is called a **solution** or **root**, and the set of all solutions is the **solution set**.

The balance rule

An equation is like a balanced scale. It stays balanced when the same number is **added to** or **subtracted from** both sides, or when both sides are **multiplied** or **divided** by the same number (never by zero).

Steps to solve (book section 7.1.2)

1. Remove brackets using the distributive property, and clear fractions by multiplying by the LCM of the denominators.
2. Transfer all terms containing the variable to **one side** and all constant terms to the **other side** - a term changes its sign when it crosses the equals sign.
3. Simplify both sides.
4. Divide both sides by the coefficient of the variable.
5. **Verify:** put the value back into the original equation. If LHS = RHS, the value is the solution.

Example 1 (book). Solve 2x + 3 = 1 - 6(x - 1).
Solution: Remove the bracket: 2x + 3 = 1 - 6x + 6, so 2x + 3 = 7 - 6x. Add 6x to both sides: 8x + 3 = 7. Subtract 3 from both sides: 8x = 4. Divide by 8: **x = 1/2**.
Check: LHS = 2(1/2) + 3 = 4. RHS = 1 - 6(1/2 - 1) = 1 - 6(-1/2) = 1 + 3 = 4. LHS = RHS, so the solution set is **{1/2}**.

2. Fractions and word problems

Example 2 (book). Solve 3x + x/5 - 5 = 1/5 + 5x.
Solution: Shift the variable terms to one side and the constants to the other: 3x + x/5 - 5x = 1/5 + 5. The left side is (x - 10x)/5 = -9x/5 and the right side is 26/5, so -9x/5 = 26/5. Multiply both sides by -5: 9x = -26. Divide by 9: **x = -26/9**.

Turning words into an equation

1. Choose a letter for the unknown quantity and write down what it stands for.
2. Express every other quantity in terms of that letter.
3. Write the given condition as an equation.
4. Solve the equation, then answer the question in words - with units.

Example 3 (book). The age of a mother is 13 times the age of her daughter. After four years it will be only five times. Find their present ages.
Solution: Let the daughter's present age be x years, so the mother's present age is 13x years. After four years: daughter x + 4, mother 13x + 4. The condition gives 13x + 4 = 5(x + 4), so 13x + 4 = 5x + 20, then 8x = 16 and **x = 2**. Present ages: **daughter 2 years, mother 13 x 2 = 26 years**. Check: after four years, 6 and 30, and 30 = 5 x 6. Correct.

Example 4 (book). A two-digit number has digits whose sum is 8. If the digits are interchanged, the new number is 36 less than the original number. Find the number.
Solution: Let the ones digit be x; then the tens digit is 8 - x. Original number = 10(8 - x) + x = 80 - 9x. Interchanged number = 10x + (8 - x) = 9x + 8. Condition: 9x + 8 = (80 - 9x) - 36, so 9x + 8 = 44 - 9x, then 18x = 36 and **x = 2**. Ones digit 2, tens digit 8 - 2 = 6, so the number is **62**. Check: 62 - 26 = 36. Correct.

3. Equations involving radicals

Definition - Radical equation: An equation in which the variable lies under a radical (square-root) sign, such as √(2x + 1) - 5 = 4, is a radical equation. Squaring both sides reduces it to a simple linear equation.

Steps to solve

1. **Isolate** the radical on one side of the equation.
2. **Square both sides** to remove the square root.
3. Solve the resulting linear equation.
4. **Verify every answer** in the original equation. A root that does not satisfy the original equation is called an **extraneous root** and is rejected.

Quick test before squaring: A square root is never negative. If isolating the radical leaves √(...) = a negative number, the equation has **no solution** at once - for example √x = -10 has solution set { }.

Example 5 (book). Find the solution set of √(3x + 2) + 6 = 2.
Solution: Isolate the radical: √(3x + 2) = 2 - 6 = -4. A square root cannot equal -4, so there can be no solution - but follow the book's working to see why checking matters. Squaring: 3x + 2 = 16, so 3x = 14 and x = 14/3. **Verify:** √(3(14/3) + 2) + 6 = √16 + 6 = 4 + 6 = 10, which is not 2. So x = 14/3 is an **extraneous root**. The equation has no solution: **solution set = { }**.

Example 6 (book Review Exercise 7). Find the solution set of √(5x + 3) + 2 = 4.
Solution: Isolate: √(5x + 3) = 2. Square both sides: 5x + 3 = 4, so 5x = 1 and **x = 1/5**. Verify: √(5(1/5) + 3) + 2 = √4 + 2 = 2 + 2 = 4. Correct, so the solution set is **{1/5}**.

4. Absolute value

Definition - Absolute value: The absolute value of a real number x, written **|x|**, is its distance from zero on the number line, so it is never negative. In symbols: |x| = x when x is zero or positive, and |x| = -x when x is negative. For example |5| = 5, |-5| = 5 and |0| = 0.

$$|\text{expression}| = k \quad (k > 0) \quad \text{means} \quad \text{expression} = k \ \text{ or } \ \text{expression} = -k$$

If k = 0, there is just one equation, expression = 0. If k is negative, there is no solution at all, because an absolute value can never be negative.

Steps to solve

1. First **isolate** the absolute-value part on one side.
2. Write the **two possibilities**: the inside equal to k, and the inside equal to -k.
3. Solve both linear equations and collect the answers as the solution set.

Example 7 (book). Solve |x - 1| = 7.
Solution: Two possibilities: (i) x - 1 = 7, giving x = 8; (ii) x - 1 = -7, giving x = -6. Solution set = **{8, -6}**.

Example 8 (book). Find the solution set of |3x - 5| + 7 = 11.
Solution: Isolate first: subtract 7 from both sides, |3x - 5| = 4. Two possibilities: 3x - 5 = 4, giving 3x = 9 and x = 3; and 3x - 5 = -4, giving 3x = 1 and x = 1/3. Solution set = **{3, 1/3}**.

Remember: An absolute-value equation normally has **two** solutions. Giving only the positive case is the most common way students lose marks in this topic.

5. Linear inequalities and the number line

> **Definition - Linear inequality:** A statement of the form ax + b > c, ax + b ≥ c, ax + b < c or ax + b ≤ c, where a, b, c are real numbers and a is not zero, is a linear inequality in one variable. The signs mean: > greater than, < less than, ≥ greater than or equal to, ≤ less than or equal to. For example 3x + 5 > 4 and 2n - 1/6 ≤ 2 are linear inequalities.

The **solution** of an inequality is any value of the variable that makes it a true statement, and the set of all such values is its **solution set**. Unlike an equation, an inequality usually has infinitely many solutions, so the answer is written as an inequality (for example x < 4) and shown on the **number line**.

Inequality	Meaning	Circle on the number line
x < 4	All real numbers less than 4; 4 itself is **not** included	**Hollow** circle at 4, arrow to the left
x ≤ 4	All real numbers less than or equal to 4; 4 **is** included	**Solid (filled)** circle at 4, arrow to the left
-2 < x < 5	All real numbers between -2 and 5; neither end is included (book Example 11)	Hollow circles at -2 and 5, segment joined between them

Properties of inequalities (book section 7.3)

- **Trichotomy property:** For any two real numbers a and b, exactly one of a < b, a = b, a > b is true.
- **Transitive property:** If a > b and b > c, then a > c. (Book example: x > 4 and 4 > z give x > z.)
- **Additive property:** Adding the same number to both sides keeps the inequality true: 21 < 31 gives 21 + 10 < 31 + 10, that is 31 < 41.
- **Multiplicative property:** Multiplying both sides by a **positive** number keeps the sign: 10 < 20 gives 30 < 60. Multiplying by a **negative** number **reverses** the sign: -12 > -15 multiplied by -2 gives 24 < 30.

6. Solving linear inequalities

Linear inequalities are solved in almost the same way as linear equations (book section 7.4), using these principles:

1. If a > b, then a + c > b + c and a - c > b - c: the same number may be added to or subtracted from both sides.
2. If a > b and k > 0, then ka > kb: both sides may be multiplied or divided by a **positive** number, sign unchanged.
3. If a > b and k < 0, then ka < kb: multiplying or dividing by a **negative** number **reverses** the inequality sign.

> **The one rule that matters most:** Multiply or divide both sides by a negative number, and the sign flips: > becomes <, and ≤ becomes ≥. Forgetting this flip is the most common error in this unit.

Example 9 (book). Solve $x - \frac{5}{7} \le \frac{(15 + 2x)}{7}$, where (i) x is a natural number, (ii) x is a real number.
Solution: Multiply both sides by 7: $7x - 5 \le 15 + 2x$. Subtract 2x: $5x - 5 \le 15$. Add 5: $5x \le 20$. Divide by 5: $\mathbf{x \le 4}$. (i) For natural numbers the solution set is **{1, 2, 3, 4}**. (ii) For real numbers it is every real x with $x \le 4$, shown on the number line with a **solid** circle at 4 and an arrow to the left.

Example 10 (book). Solve $\frac{(x + 3)}{2} \le \frac{(x - 5)}{3}$, where x is real.
Solution: Multiply both sides by 6, the LCM of 2 and 3: $3(x + 3) \le 2(x - 5)$, so $3x + 9 \le 2x - 10$. Subtract 2x and then 9: $\mathbf{x \le -19}$. On the number line: a solid circle at -19 with an arrow to the left.

Example 11. Solve $3 - 2x > 7$ and show the answer on the number line.
Solution: Subtract 3: $-2x > 4$. Divide by -2 and **reverse the sign**: $\mathbf{x < -2}$. On the number line: a hollow circle at -2 with an arrow to the left.

7. Solved examples - the book's Review Exercise 7

Example 12. Solve 5(3x + 1) = 2(x - 4).
Solution: Open the brackets: 15x + 5 = 2x - 8. Collect the variable terms: 15x - 2x = -8 - 5, so 13x = -13 and **x = -1**. Check: 5(-3 + 1) = -10 and 2(-1 - 4) = -10. Correct.

Example 13. Solve (x - 8)/3 + (x - 3)/2 = 0.
Solution: Multiply by 6, the LCM of 3 and 2: 2(x - 8) + 3(x - 3) = 0, so 2x - 16 + 3x - 9 = 0, then 5x - 25 = 0 and **x = 5**. Check: (5 - 8)/3 + (5 - 3)/2 = -1 + 1 = 0. Correct.

Example 14. Solve √(2(5x - 1)) = √(2x + 14).
Solution: Square both sides: 2(5x - 1) = 2x + 14, so 10x - 2 = 2x + 14, then 8x = 16 and **x = 2**. Check: both sides become √18. Correct, so the solution set is **{2}**.

Example 15. Solve |2x + 7| = 9.
Solution: Two possibilities: 2x + 7 = 9, giving 2x = 2 and x = 1; and 2x + 7 = -9, giving 2x = -16 and x = -8. Solution set = **{1, -8}**.

Example 16 (book, Think). A father is 4 times as old as his son. In 20 years he will be twice as old as his son. Find their present ages.
Solution: Let the son's present age be x years; the father's is 4x years. In 20 years: son x + 20, father 4x + 20. Condition: 4x + 20 = 2(x + 20), so 4x + 20 = 2x + 40, then 2x = 20 and **x = 10**. Present ages: **son 10 years, father 40 years**. Check: in 20 years, 30 and 60, and 60 = 2 x 30. Correct.

8. Double inequalities and common mistakes

A **double inequality** such as -1 < (x - 3)/2 < 0 is two inequalities at once. Do the same operation to **all three parts** at every step, and if you multiply or divide by a negative number, reverse **both** signs.

Example 17 (book Review Exercise 7). Solve -1 < (x - 3)/2 < 0 and graph the solution.
Solution: Multiply all three parts by 2: -2 < x - 3 < 0. Add 3 to all three parts: **1 < x < 3**. On the number line: hollow circles at 1 and 3, with the part between them shaded.

Example 18 (book Review Exercise 7). Solve 7 < -3x + 1 ≤ 13.
Solution: Subtract 1 from all three parts: 6 < -3x ≤ 12. Divide all three parts by -3 and reverse both signs: -2 > x ≥ -4, written in order as **-4 ≤ x < -2**. On the number line: a solid circle at -4, a hollow circle at -2, shaded between them.

Common mistakes to avoid

- Changing a term's position **without changing its sign** when it crosses the equals sign.
- Forgetting to **verify** after squaring a radical equation - an extraneous root looks like an answer but fails the check.
- Giving only **one** answer for an absolute-value equation; |...| = k has two cases.
- Dividing an inequality by a negative number and **not reversing** the sign.
- Using a **solid** circle for < or > (the end point is excluded - hollow circle), or a hollow circle for ≤ or ≥.
- In a word problem, stopping at x = 2 instead of answering the question - ages, the number itself, with units.

> **Reading the answer back:** For an equation, substitute the answer and check LHS = RHS. For an inequality, test one easy value from your solution set - for example x = 0 when the answer is x < -2 should fail, and x = -3 should work.

9. Test yourself - MCQs and short questions

Multiple-choice questions (from the book's Review Exercise 7)

1. Solve for x: (1/2)|x - 6| - 4 = -1. (a) x = 12 (b) x = 8 and x = 4 (c) x = 12 and x = 0 (d) no solution
   Answer: (c) - |x - 6| = 6, so x - 6 = 6 or x - 6 = -6.
2. Solve for x: |3x - 1| = 2. (a) x = 1 (b) x = 1 and x = -1/3 (c) x = 1 and x = -1 (d) x = 1 and x = 1/3
   Answer: (b)
3. The solution set of √(5x + 3) + 2 = 4 is: (a) {1/5} (b) {-1/5} (c) {2} (d) {1}
   Answer: (a)
4. Solve for x: √x = -10. (a) {-10} (b) { } (c) {100} (d) {10}
   Answer: (b) - a square root is never negative.
5. √(2x + 1) - 5 = 4 is a: (a) linear equation (b) radical equation (c) cubic equation (d) quadratic equation
   Answer: (b)
6. The solution of |x - 7| = 1 is: (a) x = 8 (b) x = 6 and x = 8 (c) x = 8 and x = -8 (d) x = 6
   Answer: (b)
7. The solution set of |5x/3| = 5 is: (a) {3} (b) {3, -3} (c) {5, -5} (d) {-3}
   Answer: (b) - 5x/3 = 5 or 5x/3 = -5.
8. The solution set of |-x| = 0 is: (a) {-1} (b) {1} (c) { } (d) {0}
   Answer: (d)

Short questions for written practice

1. Solve 2x + 3 = 1 - 6(x - 1). (Answer: x = 1/2)
2. Solve -1 < (x - 4)/5 < 0 and graph it. (Answer: -1 < x < 4)
3. Solve |2x + 7| = 9. (Answer: {1, -8})
4. Solve √(2(5x - 1)) = √(2x + 14). (Answer: {2})
5. Solve x - 5/7 ≤ (15 + 2x)/7 for x a natural number. (Answer: {1, 2, 3, 4})

> **One-page revision:** Equation - balance both sides, variables one side, constants the other, divide, then check. Radical - isolate, square, solve, and reject extraneous roots. Absolute value - two cases, = k and = -k. Inequality - solve like an equation, but flip the sign when multiplying or dividing by a negative; hollow circle excludes the end point, solid circle includes it.